TEXT & AUDIO SENTIMENT ANALYZER

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INTRODUCTION

- Sentiment analysis is the process of identifying and categorizing emotions, opinions, or attitudes expressed in text or audio data.
- As people increasingly share feedback through reviews, social media, and voice messages, it's important to understand this data to improve services, make better business decisions, and enhance user experience.
- This project presents a web application that can analyze both text and audio inputs for sentiment, providing results in real-time using machine learning models and lexicon-based methods.

OBJECTIVE

- Most existing sentiment analysis systems focus only on text data, leaving audio feedback largely unexplored and underutilized.
- Direct audio sentiment analysis requires complex, resource-heavy deep learning models, making them difficult to implement in small projects or real-time applications.
- There is a need for a lightweight, efficient, and easy-to-use integrated system that can analyze both text and audio inputs for sentiment in a fast and accurate manner.

TOOLS & TECHNOLOGIES USED

- Python
- Natural Language Processing (NLP) & Sentiment Analysis:
  - VADER Sentiment Analysis
  - NLTK
- Machine Learning:
  - Scikit-learn (Naïve Bayes & Logistic Regression)
- Audio Processing:
  - Librosa (Audio Feature Extraction)
- Front-end:
  - Flask

DATA OVERVIEW

Text Sentiment Dataset: "amazon_alexa" From Kaggle

rating	date	variation	verified_reviews	feedback
5	31-Jul-18	Charcoal Fabric	Love my Echo!	1
5	31-Jul-18	Charcoal Fabric	Loved it!	1
4	31-Jul-18	Walnut Finish	Sometimes while playing a game, you can answer a question corre	1
5	31-Jul-18	Charcoal Fabric	I have had a lot of fun with this thing. My 4 yr old learns about din	1
5	31-Jul-18	Charcoal Fabric	Music	1
5	31-Jul-18	Heather Gray Fabric	I received the echo as a gift. I needed another Bluetooth or some	1
3	31-Jul-18	Sandstone Fabric	Without having a cellphone, I cannot use many of her features. I h	1
5	31-Jul-18	Charcoal Fabric	I think this is the 5th one I've purchased. I'm working on getting or	1
5	30-Jul-18	Heather Gray Fabric	looks great	1

Audio Sentiment Dataset:

Custom audio clips generated using Google Text-to-Speech (gTTS).

https://drive.google.com/drive/folders/1jL2kypdz62PGRmjz6H9qnPtCeve3Q3NF?usp=drive_link

METHODOLOGY

- Text Sentiment Analysis:

Data Collection
- Amazon Alexa dataset

Preprocessing
- Tokenization, stopwords removal, cleaning

Feature Extraction
- TF-IDF Vectorization

Sentiment Classification
- Naïve Bayes
- Logistic Regression
- VADER Sentiment Analysis

METHODOLOGY

- Audio Sentiment Analysis:

Data Collection
- Custom audio clips created using gTTS

Preprocessing
- Librosa

Feature Extraction
- MFCC Extraction using Librosa

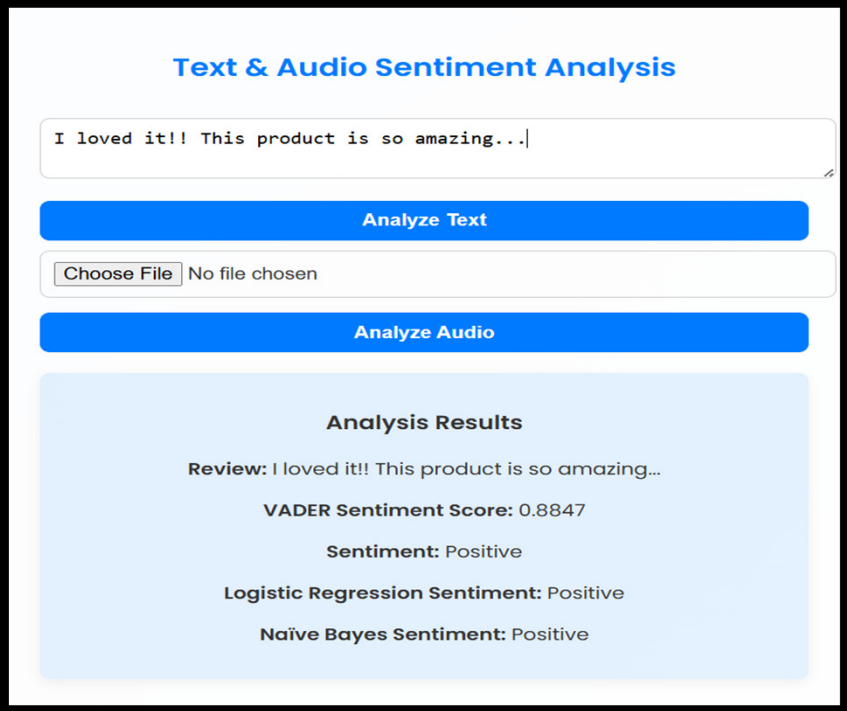
Audio Transcription
- Convert audio to text using SpeechRecognition

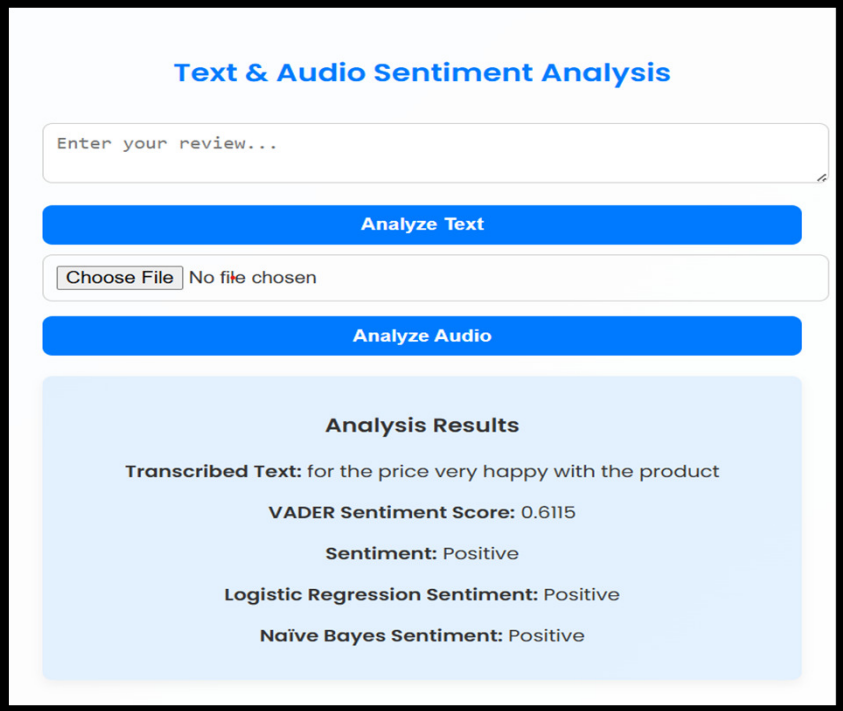
Sentiment Classification
- Naïve Bayes
- Logistic Regression
- VADER Sentiment Analysis

RESULTS & EVALUATION

- Achieved 89% accuracy with Naïve Bayes and 94% with Logistic Regression for text sentiment classification.
- Audio clips were converted to text using Google Speech Recognition, then passed through the same text sentiment analysis pipeline.
- Model performance was evaluated using Accuracy Score as the primary metric.

FINAL OUTPUT





CONCLUSION

- Successfully developed a system for text and audio sentiment analysis using machine learning and NLP techniques.
- Incorporated VADER sentiment analysis, Naïve Bayes, and Logistic Regression models for effective classification.
- Enabled a simple Flask-based web interface for users to input text or audio and receive sentiment predictions instantly.

FUTURE SCOPE

- Integrate deep learning models like LSTM or CNN for improved accuracy in audio sentiment classification.
- Expand the dataset with real-world conversational audio samples for better model generalization.
- Include a microphone for live audio input to enable real-time sentiment analysis.
- Develop a mobile or desktop application for easier access and real-time sentiment analysis.

REFERENCES

- Text sentiment dataset: https://www.kaggle.com/datasets/khaledsayedaaaaa/amazon-alexa
- Sentiment classification using ML: https://aclanthology.org/W02-1011.pdf
- Multimodel sentiment Analysis: https://arxiv.org/pdf/1803.07427

THANK YOU…